



The dual eigenvalue problems for the Sturm–Liouville system

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ABSTRACT

In this paper, we find the minimizer of the eigenvalue gap for the Schrödinger equation and vibrating string equation. In the first part, we show the first two Neumann eigenvalue gap of the Schrödinger equation with single-well potentials is not less than 1 and the equality holds if and only if the potential is constant. In the second part, since the first Neumann eigenvalue of the vibrating string equation is 0, we turn to show that the minimizing density function of the second Neumann eigenvalue is of the form $h\chi_{(a,\pi-a)} + H\chi_{[0,\pi]\setminus(a,\pi-a)}$ for some a .

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1. Introduction

The classical Sturm–Liouville problem

$$-y'' + q(x)y = \lambda\rho(x)y \quad (1)$$

has two special cases: the Schrödinger equation

$$-y'' + q(x)y = \lambda y, \quad (2)$$

and the vibrating string equation

$$-y'' = \lambda\rho(x)y \quad (3)$$

where q is the potential function and $\rho \geq \varepsilon > 0$ is the density function. Throughout this paper, we denote the eigenvalues of (1) as the following: for $k \geq 0$,

λ_k : the $(k+1)$ th periodic eigenvalue ($y_k(0) = y_k(\pi)$, $y'_k(0) = y'_k(\pi)$);

λ'_{k+1} : the $(k+1)$ th antiperiodic eigenvalue ($y_{k+1}(0) = -y_{k+1}(\pi)$, $y'_{k+1}(0) = -y'_{k+1}(\pi)$);

μ_{k+1} : the $(k+1)$ th Dirichlet eigenvalue ($y_{k+1}(0) = y_{k+1}(\pi) = 0$);

ν_k : the $(k+1)$ th Neumann eigenvalue ($y'_k(0) = y'_k(\pi) = 0$).

It is known [1,2] that

$$\begin{aligned} \nu_0 \leq \lambda_0 < \lambda'_1 \leq \frac{\mu_1}{\nu_1} \leq \lambda'_2 < \lambda_1 \leq \frac{\mu_2}{\nu_2} \leq \lambda_2 < \lambda'_3 \leq \dots \\ \leq \lambda'_{2k+1} \leq \frac{\mu_{2k+1}}{\nu_{2k+1}} \leq \lambda'_{2k+2} < \lambda_{2k+1} \leq \frac{\mu_{2k+2}}{\nu_{2k+2}} \leq \lambda_{2k+2} < \lambda'_{2k+3} \leq \dots \end{aligned} \quad (4)$$

In particular, in the vibrating string equation, $\nu_0 = \lambda_0 = 0$.

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Recently, there have been a number of studies on the optimal estimates of eigenvalues and eigenvalue gaps for the Schrödinger operator $-y'' + qy = \lambda y$ [3–5]. In particular, it was proved that the constant potential function gives the minimum Dirichlet eigenvalue gaps $\mu_2 - \mu_1$ when the potential function q is assumed to be convex [6], symmetric single-well [7] or single-well [5], while under some additional conditions, the symmetric 1-step function is the potential function in $E[h, H, M]$ giving the minimal Dirichlet eigenvalue gap [4]. Here, the function V is called a single-well function with the transition point a if $V(x)$ is decreasing in $[0, a]$ and increasing in $[a, \pi]$ while V is called a single-barrier function if $-V$ is a single-well function. And V is a symmetric single-well if V is a single-well with the transition point $\pi/2$. On the other hand, it is known that for the string equation $-y'' = \lambda \rho(x)y$, the constant density function gives the minimum Dirichlet eigenvalue ratio $\frac{\mu_2}{\mu_1}$ when the density function ρ is assumed to be concave, symmetric single-barrier [8] or single-barrier [5], while the symmetric 1-step function is the density in $E[h, H, M]$ giving the minimum Dirichlet eigenvalue ratio [9]. See also [10]. These results are called “duality results”. In particular, Huang in 2007 discussed the eigenvalue gap for vibrating string with symmetric single-well densities [11].

In [5], Horvath considered the Dirichlet Sturm–Liouville operator and showed that $\mu_2 - \mu_1 \geq 3$ for (2) with single-well potentials and $\frac{\mu_2}{\mu_1} \geq 4$ for (3) with single-barrier densities. Inspired by the interesting results, we cannot but ask *whether or not we can obtain a similar result for the Neumann string equation?* For the Neumann Schrödinger equation, we have a similar inequality (cf. Theorem 1.1). But for the Neumann string equation, since $v_0 = 0$, we will discuss $\min v_1$ (cf. Theorem 1.5).

Theorem 1.1. Consider (2) with Neumann boundary conditions. Let q be a single-well potential on $[0, \pi]$ with transition point $a = \frac{\pi}{2}$. Then

$$v_1 - v_0 \geq 1,$$

and the equality holds if and only if q is constant. If $a \neq \frac{\pi}{2}$, then there are single-well potentials q such that $v_1 - v_0 < 1$.

In 1992, Ashbaugh and Svirsky [3] studied the gap $\lambda'_1 - \lambda_0$ between first antiperiodic eigenvalue and first periodic eigenvalue for L^∞ potential functions. Let

$$\begin{aligned}\tilde{S}(M) &= \{q \in L^\infty(-\infty, \infty) | q(x+1) = q(x) \text{ and } \|q\|_\infty \leq M\}, \\ S(M) &= \{q \in L^\infty(0, \pi) | \|q\|_\infty \leq M\}.\end{aligned}$$

Theorem 1.2 ([3]).

(a) $(\lambda'_1 - \lambda_0)(q_*) = \min_{\tilde{S}(M)} (\lambda'_1 - \lambda_0)(q)$, where potential function

$$q_* = \begin{cases} M, & \text{on } B^+ = \{x \in \mathbb{R} | u_0^2 > u_1^2\}. \\ -M, & \text{on } B^- = \{x \in \mathbb{R} | u_0^2 < u_1^2\}. \end{cases}$$

(b) $\min_{\tilde{S}(M)} (\lambda'_1 - \lambda_0) = \min_{S(M)} (\mu_1 - v_0) = \min_{S(M)} (v_1 - v_0)$.

It would be desirable to study the dual problem of minimum eigenvalue ratio $\frac{\lambda'_1}{\lambda_0}$ for the string equation. However for the string equation $\lambda_0 = 0$, so instead we investigate the problem of $\min \lambda'_1 = \min(\lambda'_1 - \lambda_0)$ for the string equation. Our Theorems 1.4 and 1.5 are a bit similar to that of Chern–Shen.

Theorem 1.3 ([4]). For $0 < h \leq H$, $h\pi \leq M \leq H\pi$, define

$$E[h, H, M] = \left\{ q \in L^\infty(0, \pi) : h \leq q \leq H \text{ a.e. and } \int_0^\pi q(x) dx = M \right\}.$$

Then the potential function $q_* \in E[h, H, M]$ giving the minimum Dirichlet eigenvalue μ_1 is $h\chi_{[0,1] \setminus (a,b)} + H\chi_{(a,b)}$.

Similarly, for $0 < h \leq H$, $h\pi \leq M \leq H\pi$, define

$$\tilde{E}[h, H, M] = \left\{ \rho \in L^\infty(-\infty, \infty) : h \leq \rho \leq H \text{ a.e., } \rho(x+\pi) = \rho(x) \text{ and } \int_0^\pi \rho(x) dx = M \right\}.$$

We remark that $E[h, H, M]$ is isometric to $\tilde{E}[h, H, M]$ by a simple extension of ρ , because functions in $\tilde{E}[h, H, M]$ can be viewed as periodic extensions of the function in $E[h, H, M]$.

Theorem 1.4. Let ρ_0 be a minimizing (maximizing) density function of λ'_1 in $\tilde{E}[h, H, M]$. Then up to a translation,

$$\rho_0 = h\chi_{(a,b)} + H\chi_{[0,\pi] \setminus (a,b)} \quad (\rho_0 = H\chi_{(a,b)} + h\chi_{[0,\pi] \setminus (a,b)}) \quad \text{a.e.}$$

where $0 < a < b < \pi$, and $b - a = \frac{H\pi - M}{H - h}$ ($b - a = \frac{M - h\pi}{H - h}$).

Theorem 1.5. Consider the vibrating string equation. Then

$$\min_E \lambda'_1 = \min_E \mu_1 = \min_E \nu_1,$$

and the minimizing density function of ν_1 is of the form

$$h\chi_{(a, \pi-a)} + H\chi_{[0, \pi] \setminus (a, \pi-a)}.$$

According to (4), it suffices to show that $\min_E \lambda'_1 \geq \min_E \mu_1, \min_E \nu_1$ in Theorem 1.5. The proofs of Theorems 1.4 and 1.5 will be given in Section 3.

2. The first Neumann eigenvalue gap of the Schrödinger operator

Let $(\nu_n, y_n)_{n \geq 0}$ be the $(n+1)$ th normalized eigenpair of (2) with Neumann boundary conditions satisfying $y_n(x) > 0$ for small $x > 0$. In this section, we will discuss the Neumann eigenvalue gap of the Schrödinger operator with the single-well potentials. The technique comes from [5].

Let $q(x, t)$ be a one-parameter family of potential functions such that $\frac{\partial q}{\partial t}(x, t)$ exists.

Lemma 2.1 ([6]).

$$\frac{d}{dt} \nu_n(t) = \int_0^\pi \frac{\partial q}{\partial t}(x, t) y_n^2(x, t) dx.$$

Lemma 2.2. The equation $|y_0(x)| = |y_1(x)|$ has at most two solutions in $(0, \pi)$.

Proof. Let x_0 be the zero of y_1 in $(0, \pi)$. Suppose there exist $\alpha_1, \alpha_2 \in (0, x_0)$ such that

$$|y_0(\alpha_i)| = |y_1(\alpha_i)|, \quad i = 1, 2.$$

Define $v(x) = \frac{y_1(x)}{y_0(x)}$. Then $v(\alpha_1) = v(\alpha_2) = 1$. By Rolle's Theorem, there is $\xi \in (\alpha_1, \alpha_2)$ such that $v'(\xi) = 0$.

On the other hand, for all $x \in (0, x_0)$,

$$\begin{aligned} v'(x) &= \frac{y_1'(x)y_0(x) - y_0'(x)y_1(x)}{y_0^2(x)} \\ &= -\frac{1}{y_0^2(x)} \int_0^x (v_1 - v_0)y_0(t)y_1(t) dt \\ &< 0. \end{aligned}$$

It is a contradiction. Hence there is at most one zero of $|y_0(x)| = |y_1(x)|$ in $(0, x_0)$. The case of the interval (x_0, π) is similar. $\square$

By Lemma 2.2, there exist $0 \leq x_- < x_0 < x_+ \leq \pi$ such that $y_1(x_0) = 0$ and

$$y_1^2 - y_0^2 \begin{cases} > 0 & \text{on } (0, x_-) \cup (x_+, \pi), \\ < 0 & \text{on } (x_-, x_+). \end{cases} \quad (5)$$

Here, if $|y_0(x)|$ and $|y_1(x)|$ have no intersection in $(0, x_0) \cup (x_0, \pi)$, then we assume $x_- = 0$ ($x_+ = \pi$).

Lemma 2.3. Define $g(t) = \sqrt{t} \tan(\sqrt{t} \frac{\pi}{2})$ for real t and let $m > 0$. Then the first two real solutions of the equation $g(t) = -g(t-m)$ satisfy

$$t_2 - t_1 > 1.$$

Proof. (i) Since

$$\begin{aligned} g'(t) &= \frac{\sin(\sqrt{t}\pi) + \sqrt{t}\pi}{4\sqrt{t} \cos^2(\sqrt{t} \frac{\pi}{2})} > 0 \quad \text{for } t > 0, t \neq (2n+1)^2, \\ g'(t) &= \frac{\sinh(\sqrt{-t}\pi) + \pi\sqrt{-t}}{4\sqrt{-t} \cosh^2(\sqrt{-t} \frac{\pi}{2})} > 0 \quad \text{for } t < 0, \end{aligned}$$

the function $g(t)$ is strictly increasing in the intervals $(-\infty, 1)$, $(1, 9)$ and in general in $((2n-1)^2, (2n+1)^2)$, $n \geq 2$. Moreover,

$$t_1 \in (0, 1), \quad t_2 \in (1, 9), \quad g(t_1) > 0.$$

(ii) Let $t(m)$ denote the solution of $g(t) = -g(t - m)$. Then

$$g'(t) \frac{dt}{dm} = -g'(t - m) \left(\frac{dt}{dm} - 1 \right), \quad (6)$$

and this is equivalent to

$$(g'(t) + g'(t - m)) \frac{dt}{dm} = g'(t - m).$$

By (i), we obtain $\frac{dt}{dm} > 0$. That is, $t_1(m)$ and $t_2(m)$ are strictly increasing functions of $m > 0$.

(iii) When $m = 4$, $t_1 \in (0, 1)$, $t_2 = 4$. Since $t_2(m)$ is strictly increasing in m , we obtain

$$t_2 - t_1 > 4 - 1 = 3 \quad \text{for } m \geq 4.$$

Hence, we only need to consider $0 < m < 4$. In this case, $1 < t_2 < 4$ and $g(t_2) < 0$. Note that $t_1 < m < t_2$.

(iv) Suppose there exists m^* such that $2t_1(m^*) > m^*$. Since $g(t)$ is strictly increasing for $t < 0$, the inequality $-t_1 < t_1 - m^*$ is equivalent to

$$-\frac{\sqrt{t_1} \sinh(\sqrt{t_1} \frac{\pi}{2})}{\cosh(\sqrt{t_1} \frac{\pi}{2})} = g(-t_1) \leq g(t_1 - m^*) = -g(t_1),$$

and hence,

$$\tanh\left(\sqrt{t_1} \frac{\pi}{2}\right) \geq \tan\left(\sqrt{t_1} \frac{\pi}{2}\right).$$

But it is obvious that $\tanh s < \tan s$ for $0 < s < \frac{\pi}{2}$. Hence $2t_1(m^*) < m^*$.

(v) Suppose t^* is a solution of $g(t) = -g(t - m)$. Since $g'(t) = \frac{1}{2\sqrt{t}} \tan\left(\sqrt{t} \frac{\pi}{2}\right) + \frac{\pi}{4} [1 + \tan^2\left(\sqrt{t} \frac{\pi}{2}\right)]$, we have

$$g'(t^* - m) - g'(t^*) = -\frac{\tan\left(\sqrt{t^*} \frac{\pi}{2}\right)}{4\sqrt{t^*}(t^* - m)} [4t^* - 2m - m\pi g(t^*)].$$

For $t^* = t_2$, since $m < t_2$, $1 < t_2 < 4$ and $g(t_2) < 0$, we obtain $g'(t_2 - m) > g'(t_2)$.

For $t^* = t_1$, since $t_1 < 2t_1 < m$ and $g(t_1) > 0$, we obtain $g'(t_1 - m) < g'(t_1)$.

(vi) By (i), (v), and (6), we obtain

$$\frac{1 - t'_2(m)}{t'_2(m)} = \frac{g'(t_2(m))}{g'(t_2(m) - m)} < 1 < \frac{g'(t_1(m))}{g'(t_1(m) - m)} = \frac{1 - t'_1(m)}{t'_1(m)}.$$

Hence, $t'_1(m) < \frac{1}{2} < t'_2(m)$, and then

$$\frac{d}{dm}(t_2(m) - t_1(m)) > 0, \quad 0 < m < 4.$$

Since $\lim_{m \rightarrow 0} (t_2(m) - t_1(m)) = 1$, we obtain

$$t_2(m) - t_1(m) > 1. \quad \square$$

Proof of Theorem 1.1. For $M > 0$, denote

$$A_M = \left\{ q : 0 \leq q \leq M, q \text{ is a single-well potential with a transition point at } \frac{\pi}{2} \right\}.$$

It is clear that A_M is closed, and then there exists an optimal potential $q_0 \in A_M$ giving the minimal value $v_1 - v_0$ over A_M . Furthermore, by Lemma 2.2, there are $0 \leq x_- < x_+ \leq \pi$ corresponding to q_0 satisfying (5).

First, consider $x_- \leq \frac{\pi}{2} \leq x_+$. Let

$$q_1(x) = \begin{cases} q_0(x_-) & \text{on } \left(0, \frac{\pi}{2}\right), \\ q_0(x_+) & \text{on } \left(\frac{\pi}{2}, \pi\right), \end{cases}$$

and define $q(x, t) = tq_1(x) + (1 - t)q_0(x)$ for $t \in [0, 1]$. Then $q(x, t) \in A_M$ and

$$q_1 - q_0 \begin{cases} \leq 0 & \text{on } (0, x_-) \cup (x_+, \pi), \\ \geq 0 & \text{on } (x_-, x_+). \end{cases} \quad (7)$$

By the optimality of q_0 , we have

$$0 \leq \frac{d}{dt}(v_1(t) - v_0(t))|_{t=0} = \int_0^\pi (q_1(x) - q_0(x))(y_1^2(x, 0) - y_0^2(x, 0))dx.$$

By (5) and (7), the above integral equals zero and hence $q_0 = q_1$ a.e. on $[0, \pi]$.

If $x_+ < \frac{\pi}{2}$. Let

$$q_1(x) = \begin{cases} M & \text{on } (0, x_+), \\ 0 & \text{on } (x_+, \pi). \end{cases}$$

By the definition of x_+ and $\int_0^\pi y_n^2(x) dx = 1$, we obtain

$$\int_{x_+}^\pi (y_1^2(x, 0) - y_0^2(x, 0)) dx > 0 > \int_0^{x_+} (y_1^2(x, 0) - y_0^2(x, 0)) dx.$$

This gives by the optimality of q_0 that

$$\begin{aligned} 0 &\leq \frac{d}{dt} (v_1(0) - v_0(0)) \\ &= \int_0^\pi (q_1(x) - q_0(x)) (y_1^2(x, 0) - y_0^2(x, 0)) dx \\ &\leq (M - q_0(x_-)) \int_0^{x_+} (y_1^2(x, 0) - y_0^2(x, 0)) dx - q_0\left(\frac{\pi}{2}\right) \int_{x_+}^\pi (y_1^2(x, 0) - y_0^2(x, 0)) dx \\ &\leq 0. \end{aligned}$$

Hence $q_0(x) = M$ on $(0, x_+)$ and 0 on (x_+, π) . However, the second eigenfunction of this potential q_0 can be expressed by

$$y_1(x) = \begin{cases} c \cos(\sqrt{v_1 - M}x) & \text{on } (0, x_+), \\ d \cos(\sqrt{v_1}(\pi - x)) & \text{on } (x_+, \pi), \end{cases}$$

where c and d are constants. Since the only zero $x_0 \in (x_-, x_+) \subset (0, \frac{\pi}{2})$, we obtain $\sqrt{v_1 - M}x_+ > \frac{\pi}{2}$ and $\sqrt{v_1}\frac{\pi}{2} < \frac{\pi}{2}$, i.e. $1 < v_1 - M < 1$. Thus the possibility that $x_+ < \frac{\pi}{2}$ is refuted.

The case is similar when $x_- > \frac{\pi}{2}$.

After simplification, the optimal potential q_0 must be a 1-step function. W.L.O.G. let

$$q_0 = \begin{cases} m & \text{on } (0, \frac{\pi}{2}), \\ 0 & \text{on } (\frac{\pi}{2}, \pi), \end{cases}$$

where $m \geq 0$. Under this potential, an eigenfunction corresponding to the eigenvalue can be expressed as

$$y(x) = \begin{cases} c \cos(\sqrt{\lambda - m}x) & \text{on } (0, \frac{\pi}{2}), \\ d \cos(\sqrt{\lambda}(\pi - x)) & \text{on } (\frac{\pi}{2}, \pi). \end{cases}$$

Since $y(x)$ is C^1 at the jump $\frac{\pi}{2}$, it must satisfy

$$\lim_{x \uparrow \frac{\pi}{2}} \frac{y'(x)}{y(x)} = \lim_{x \downarrow \frac{\pi}{2}} \frac{y'(x)}{y(x)}$$

i.e.

$$\sqrt{\lambda} \tan\left(\sqrt{\lambda} \frac{\pi}{2}\right) = -\sqrt{\lambda - m} \tan\left(\sqrt{\lambda - m} \frac{\pi}{2}\right). \quad (8)$$

Note that the eigenvalues are the real solutions of (8). By Lemma 2.3, we know $v_1 - v_0 > 1$ if $m > 0$. Hence the optimal $m = 0$ and q_0 must be constant a.e.

Finally, let

$$q(x, t) = \begin{cases} t & \text{if } x \in (0, a), \\ 0 & \text{if } x \in (a, \pi), \end{cases}$$

Since $y_0(x, 0) = \sqrt{\frac{1}{\pi}}$, $y_1(x, 0) = \sqrt{\frac{2}{\pi}} \cos x$, $\int_0^{\frac{\pi}{2}} (y_1^2(x, 0) - y_0^2(x, 0)) dx = 0$ and $y_1^2(x, 0) < y_0^2(x, 0)$ on $(\frac{\pi}{3}, \frac{2\pi}{3})$, we have

$$\frac{d}{dt} (v_1(0) - v_0(0)) = \int_0^a (y_1^2(x, 0) - y_0^2(x, 0)) dx < 0$$

when $\frac{\pi}{2} < a \leq \frac{2\pi}{3}$. That is if $\frac{\pi}{2} < a \leq \frac{2\pi}{3}$, then $q(x, t)$ for small $t > 0$ gives an eigenvalue gap $v_1 - v_0 < 1$. $\square$

3. Extremal antiperiodic eigenvalues of the string equation

In this section, we will prove Theorems 1.4 and 1.5. We shall employ the method of variational analysis, as used in previous literature. Let $\rho = \rho(x, t)$ to depend smoothly on another variable t . Hence all the parameters in (3), y and λ also depend on t .

Lemma 3.1 ([8]). *Let λ be any Dirichlet, Neumann, periodic or antiperiodic eigenvalue for (3). Then*

$$\frac{d}{dt}\lambda(t) = -\lambda(t) \int_0^\pi \frac{\partial \rho}{\partial t}(x, t) y^2(x, t) dx \quad (9)$$

where y is the eigenfunction corresponding to λ .

Consider $-y'' = \lambda \rho y$ on $(0, \pi)$ under any separated boundary condition or periodic boundary condition. Let $y_n(x)$, $n \geq 1$, be the n th eigenfunction of the above problem, then $y_n(x)$ has $n - 1$ simple zero $\{x_k^{(n)} : k = 1, 2, \dots, n - 1\}$ in $(0, \pi)$. Let $x_0^{(n)} = 0$, $x_n^{(n)} = \pi$. In between each interval $(x_k^{(n)}, x_{k+1}^{(n)})$, $k = 1, 2, \dots, n - 2$, $y(x)$ is either positive or negative. So choose a relative minimum or relative maximum point and call it $z_k^{(n)}$, $k = 1, 2, \dots, n - 2$. In this way, we have $n - 2$ relative extremal points $\{z_k^{(n)} : k = 1, 2, \dots, n - 2\}$ where each $y'(z_k^{(n)}) = 0$ if $z_k^{(n)}$ is an interior point. If there exists an additional extremal point in $(0, x_1^{(n)})$, then we call this extremal point $z_0^{(n)}$, and let $z_{-1}^{(n)} = 0$. If there is no such extremal point, we let $z_0^{(n)} = 0$. Similarly, if there is an additional extremal point in $(x_{n-1}^{(n)}, \pi)$, we call this point $z_{n-1}^{(n)}$ and let $z_n^{(n)} = \pi$. If there is no such point, then we let $z_{n-1}^{(n)} = \pi$. Let J_n be the collection of all $z_k^{(n)}$ defined above. Let $\sigma_n = \{k : z_k^{(n)} \in J_n\}$, $I_k^{(n)} = (z_k^{(n)}, z_{k+1}^{(n)}) \subseteq [0, \pi]$. So for all n , $\{0, 1, \dots, n - 1\} \subseteq \sigma_n \subseteq \{-1, 0, 1, \dots, n\}$ and σ_n has at most $n + 2$ elements.

Lemma 3.2. y_n is strictly monotonic in each subinterval $I_k = (z_k^{(n)}, z_{k+1}^{(n)})$, for all $k \in \sigma_n$.

Proof. There are three cases:

- (i) $1 \leq k \leq n - 2$, then there exists a unique nodal point $x_{k+1}^{(n)} \in (z_k^{(n)}, z_{k+1}^{(n)})$. Without loss of generality, let $y \geq 0$ on $(z_k^{(n)}, x_{k+1}^{(n)}) = I_k^-$. So if $x \in I_k^-$, then

$$-y'(x) + y'(z_k^{(n)}) = - \int_{z_k^{(n)}}^x y''(t) dt = \int_{z_k^{(n)}}^x \lambda \rho(t) y(t) dt > 0.$$

Thus $y'(x) < 0$. Hence y is strictly decreasing on $(z_k^{(n)}, x_{k+1}^{(n)})$. If $x \in (x_{k+1}^{(n)}, z_{k+1}^{(n)}) = I_k^+$, then $y < 0$ on I_k^+ . Therefore

$$y'(x) = - \int_x^{z_{k+1}^{(n)}} y''(t) dt = \int_x^{z_{k+1}^{(n)}} \lambda \rho(t) y(t) dt < 0.$$

Thus y is strictly decreasing on $I_k = (z_k^{(n)}, z_{k+1}^{(n)})$.

- (ii) $k = 0$ or $n - 1$, and there exists a nodal point in I_k . Without loss of generality, let $k = 0$, then $x_1^{(n)} \in I_0 = (z_0^{(n)}, z_1^{(n)})$, and let $y > 0$ on $I_0^- = (z_0^{(n)}, x_1^{(n)})$. So $y < 0$ on $I_0^+ = (x_1^{(n)}, z_1^{(n)})$. For all $x \in I_0^+ \cup \{x_1^{(n)}\}$,

$$y'(x) = - \int_x^{z_1^{(n)}} y''(t) dt = \int_x^{z_1^{(n)}} \lambda \rho(t) y(t) dt < 0.$$

Also as $z_0^{(n)}$ is a relative maximum, $y'(z_0^{(n)}) \leq 0$. Therefore, for all $x \in I_0^-$,

$$-y'(x) = - \int_{z_0^{(n)}}^x y''(t) dt = \int_{z_0^{(n)}}^x \lambda \rho(t) y(t) dt > 0.$$

So y is strictly decreasing on $(z_0^{(n)}, z_1^{(n)})$. Similarly, if $k = n - 1$, y is strictly decreasing on $(z_{n-1}^{(n)}, z_n^{(n)})$.

- (iii) $k = -1$ or $k = n$. Suppose $-1 \in \sigma_n$, then $z_{-1}^{(n)} = 0$ and $y > 0$ on $I_{-1} = (0, z_0^{(n)})$. So for all $x \in I_{-1} \cup \{0\}$,

$$y'(x) = - \int_x^{z_0^{(n)}} y''(t) dt = \int_x^{z_0^{(n)}} \lambda \rho(t) y(t) dt > 0.$$

Hence y is strictly increasing on I_{-1} . $\square$

Proof of Theorem 1.4. By Lemma 3.2, let α be the critical point of y_1 , we know that y_1 is strictly decreasing in $(0, \alpha)$ and then strictly increasing in (α, π) . Suppose $h < \rho < H$ on a subset A of $(0, \pi)$ with positive measure. Then

$$A = \bigcup_{n=2}^{\infty} A_n$$

where $A_n = \{x \in (0, \pi) : h + \frac{1}{n} < \rho < H - \frac{1}{n}\}$.

So there exists n_0 such that $m(A_{n_0}) > 0$. W.L.O.G. we let $B = A_{n_0} \cap (0, \alpha)$ such that $m(B) \neq 0$. Let $\beta_1, \beta_2 \in (0, \alpha)$ be such that

$$\begin{aligned} B_1 &= (0, \beta_1) \cap B, \\ B_2 &= (\beta_1, \beta_2) \cap B, \\ B_3 &= (\beta_2, \alpha) \cap B, \end{aligned}$$

such that $m(B_1) = m(B_2) = m(B_3) = c_0$, for some positive constant c_0 . Define $P \in \tilde{E}[h, H, M]$ such that $P|_{[0, \pi]} = \frac{1}{n_0}(\chi_{B_1} - \chi_{B_3})$, $\rho = \rho_0 + tP \in \tilde{E}[h, H, M]$. Then

$$\begin{aligned} 0 &\leq \frac{d}{dt} \lambda'_1(t)|_{t=0} \\ &= -\lambda'_1 \int_0^\pi P(x) y_1(x)^2 dx \\ &= -\frac{\lambda'_1}{n_0} \left[\int_{B_1} y_1(x)^2 dx - \int_{B_3} y_1(x)^2 dx \right] \\ &\leq -\frac{\lambda'_1}{n_0} c_0 [y_1(\beta_1)^2 - y_1(\beta_2)^2]. \end{aligned}$$

It leads to a contradiction. Similarly, we can eliminate the possibility that $h < \rho < H$ on any subset of (α, π) with positive measure. Therefore, $\rho_0 \in \{h, H\}$ for a.e. $x \in (0, \pi)$. So let

$$G = \{x \in (0, \pi) : \rho_0 = h\}.$$

Then

$$\rho_0(x) = \begin{cases} h, & x \in G, \\ H, & x \in (0, \pi) \setminus G. \end{cases}$$

and

$$h \cdot m(G) + H[\pi - m(G)] = M$$

so that

$$m(G) = \frac{H\pi - M}{H - h}.$$

Finally we claim that $G = (a, b)$ for some interval $\alpha \in (a, b) \subseteq (0, \pi)$. In fact, for $t \geq 0$, let

$$S_t = \{x \in (0, \pi) : |y_1(x)| < t\}.$$

Define $f : [0, \infty) \rightarrow [0, \pi]$ by $f(t) = m(S_t)$. Obviously, f is a continuous function. Hence if $r = \frac{H\pi - M}{H - h}$, then $0 \leq r \leq \pi$ and there is $t_0 \geq 0$ such that $f(t_0) = r$. Note that $\alpha \in S_{t_0}$. Also $S_{t_0} = (a, b)$ for some $0 \leq a < b \leq \pi$. Finally, if $\rho_0 = H$ on some subset $\bar{B}$ of S_{t_0} with positive measure, then there is some set $\tilde{B}$ in $(0, \pi) \setminus S_{t_0}$ with the same measure on which $\rho_0 = h$. Let $P \in \tilde{E}[h, H, M]$ such that $P|_{(0, \pi)} = (H - h)(\chi_{\tilde{B}} - \chi_{\bar{B}})$. Then we may repeat the previous argument to arrive at a contradiction. Therefore $G = S_{t_0} = (a, b)$.

The proof for maximizing density function is similar. $\square$

Lemma 3.3. Let φ_1 be the second Neumann eigenfunction with the zero x_0 . Suppose $\rho(x_0 + x) \geq \rho(x_0 - x)$ for $x \in [0, \min\{x_0, \pi - x_0\}]$. Then

- (a) $x_0 \in [\frac{\pi}{2}, \pi)$.
- (b) $\varphi_1^2(x_0 + x) \leq \varphi_1^2(x_0 - x)$ for $x \in [0, \pi - x_0]$.

Proof. Let $Q(x) = v_1 \rho(x)$. Define $z_1(x) = -\varphi_1(x_0 + x)$, $Q_1(x) = Q(x_0 + x)$ and $z_2(x) = \varphi_1(x_0 - x)$, $Q_2(x) = Q(x_0 - x)$. Note that $z'_1(0) = z'_2(0)$.

(a) Compare the following two problems

$$\begin{cases} z''_1(x) + Q_1(x)z_1(x) = 0 & \text{on } [0, \pi - x_0], \\ z_1(0) = 0, z'_1(\pi - x_0) = 0, \end{cases}$$

and

$$\begin{cases} z''_2(x) + Q_2(x)z_2(x) = 0 & \text{on } [0, x_0], \\ z_2(0) = 0, z'_2(x_0) = 0. \end{cases}$$

Since $Q_2(x) \leq Q_1(x)$ on $[0, \min\{x_0, \pi - x_0\}]$, we have $x_0 \geq \pi - x_0$ or equivalent to $x_0 \geq \frac{\pi}{2}$.

(b) Define $w(x) = \frac{z_1(x)}{z_2(x)}$ and $\hat{w}(x) = z_1'(x)z_2(x) - z_2'(x)z_1(x)$. Then we have $\hat{w}(0) = 0$ and $\hat{w}'(x) = z_1''(x)z_2(x) - z_2''(x)z_1(x) = (Q_2(x) - Q_1(x))z_1(x)z_2(x) < 0$ on $(0, \pi - x_0]$. Hence $\hat{w}(x) < 0$ on $(0, \pi - x_0]$. This implies $w'(x) = \frac{\hat{w}(x)}{z_2^2(x)} < 0$ on $(0, \pi - x_0]$. Moreover, since $w(0) = 1$, we have $w(x) < 1$ on $(0, \pi - x_0]$ and then $z_1(x) < z_2(x)$ on $(0, \pi - x_0]$. This completes the proof. $\square$

Proof of Theorem 1.5. We claim that $\min_{\bar{E}} \lambda_1' \geq \min_E \mu_1$. Let $\bar{\lambda}_1' = \min_{\bar{E}} \lambda_1'$, and ρ_0 be the density function corresponding to $\bar{\lambda}_1'$. Denote φ_1 as the eigenfunction associated with $(\bar{\lambda}_1', \rho_0)$.

If $\varphi_1(0) = -\varphi_1(\pi) = 0$, then $\bar{\lambda}_1'$ is obviously the first eigenvalue of the Dirichlet string equation $-y'' = \lambda \rho_0(x)y$ and hence

$$\min_{\bar{E}} \lambda_1' = \bar{\lambda}_1' \geq \min_E \mu_1.$$

Otherwise, let $\bar{\varphi}_1(x) = \varphi_1(x + x_0)$ and $\bar{\rho}(x) = \rho_0(x + x_0)$, $\bar{\lambda}_1'$ is still the first eigenvalue corresponding to the eigenfunction $\bar{\varphi}_1(x)$ of the Dirichlet string equation $-y'' = \lambda \bar{\rho}(x)y$. Hence, we also have $\min_{\bar{E}} \lambda_1' \geq \min_E \mu_1$.

Similarly, we can translate ρ_0 to a new density $\hat{\rho}$ such that the corresponding eigenfunction $\hat{\varphi}_1(x)$ satisfies the Neumann boundary conditions. In this case, $\bar{\lambda}_1'$ is the second eigenvalue of the Neumann string equation $-y'' = \lambda \hat{\rho}(x)y$. Hence

$$\min_{\bar{E}} \lambda_1' = \bar{\lambda}_1' \geq \min_E \nu_1.$$

In the following, we will show that the minimizing density function of ν_1 is of the form

$$h\chi_{(a, \pi-a)} + H\chi_{[0, \pi] \setminus (a, \pi-a)}.$$

Using an argument similar to that of Theorem 1.4, it can be shown that the minimizing density function $\rho_0(x)$ of ν_1 is of the form

$$h\chi_{(a, b)} + H\chi_{[0, \pi] \setminus (a, b)},$$

for some $0 < a < b < \pi$.

Suppose $a + b \neq \pi$. W.L.O.G. let $a < \pi - b$. If $b \leq x_0$, then $\varphi_1(a + \frac{a+\pi-b}{2}) > \varphi_1(b - \frac{a+\pi-b}{2})$. If $b > x_0$, then $\varphi_1(a + \frac{a+\pi-b}{2}) > \varphi_1(2x_0 - b - \frac{a+\pi-b}{2}) > |\varphi_1(b + \frac{a+\pi-b}{2})|$. Hence, using the same argument as that of Theorem 1.4, it can be shown that ρ_0 is not a minimizing density function. $\square$

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